

Dual ordinal spectral ranking of journals and institutions

DOCAMPO, D. AND CRAM, L.

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Abstract

We develop a dual spectral ranking of journals and institutions that jointly determines prestige scores via a Katz resolvent applied to a block attention matrix constructed from intra-field citation flows. A continuous parameter χ apportions each reference between source and institution channels; a binary block-mask selects among four bibliometrically distinct flow structures, nesting conventional single-mode rankings as special cases. Applied to the economics and commerce literature using OpenAlex data, the method produces stable, interpretable joint rankings whose sensitivity to model parameters is examined systematically.

Keywords: Scientometrics, spectral ranking, dual journal–institution prestige, Katz resolvent, Pinski–Narin

1 Introduction

Journal and institution ranking. The research system relies on journals and institutions. Journals filter works through editorial and peer review, archive the literature, and create metadata for retrieval. Institutions fund individual researchers, implement training, oversee compliance, and maintain physical and information infrastructure. [Placeholder: overview of the joint ranking problem.]

Previous work on journal ranking. ? introduced influence-per-reference weights for journal ranking; ? surveys the broader lineage; ? traces the Katz–Hubbell–Pinski–Narin–PageRank chain. [Placeholder: expand journal ranking literature.]

How institutions are ranked. Institutional rankings, dominated by QS, Times Higher Education (THE), and ShanghaiRanking (ARWU), blend bibliometrics with reputational surveys, teaching metrics, and internationalisation. The Leiden Ranking counters with field-normalised bibliometrics, revealing niche leaders but lacking the reputational component of QS and THE. Spectral methods have not previously been applied jointly to sources and institutions. [Placeholder: expand institutional ranking literature.]

Theoretical framing. We treat citations as structural relations within a functionally differentiated scientific system (?), setting aside authorial intent. Dual spectral rankings of journals and institutions treat citations as autopoietic communications within science’s communicative system (Social System of Science Citation Theory, SSCT), where individual actors reside in the environment rather than constituting system elements (?). This contrasts with actor-network theory (ANT), which foregrounds heterogeneous agents in assembling citation networks; ANT enters only in the Discussion. The resulting dual rankings are self-referential hierarchies that condition future citations. Our construction extends ?: all four blocks of the joint attention matrix are primary flows, not derived projections.

Paper structure. Section ?? develops the mathematical framework. Section ?? describes the corpus construction and source selection. Section ?? presents results. Section ?? discusses validity and prospects.

2 Journal–institution citation networks

Corpus construction. Following ?, a *reference* written by a scientist becomes a *citation*—a directed edge (w_i, w_j) —only after indexer resolution to canonical work records. The corpus is built in five steps: (i) select

N_s sources $\mathfrak{S}$ by field classification; (ii) collect all works $\mathfrak{W}$ ($|\mathfrak{W}| = N_w$) in $\mathfrak{S}$ within the census window $[t_0^c, t_1^c]$; (iii) retain references from $\mathfrak{W}$ to works $\mathfrak{W}^t$ in $\mathfrak{S}$ within the target window $[t_0^t, t_1^t]$, forming $\mathcal{R} = \{(i, j) : w_i \in \mathfrak{W}, w_j \in \mathfrak{W}^t\}$; (iv) induce institution set $\mathfrak{I}'$ from author affiliations of $\mathfrak{W}$; (v) retain institutions with $\geq \tau$ census works, giving $\mathfrak{I}$, $|\mathfrak{I}| = N_u$. Flows crossing any corpus boundary are excluded without renormalisation (*open-corpus principle*); the closed-corpus alternative is examined in the Supplement.

Each work w_i is assigned an institution weight under author-fractional counting,

$$\omega_{ik} = \sum_{j: k \in \mathcal{U}_{ij}} \frac{1}{A_i} \frac{1}{U_{ij}}, \quad (1)$$

where A_i is the number of authors and U_{ij} the number of retained institutions of author j ; after filtering, $\sum_k \omega_{ik} \leq 1$ and weights are not renormalised. Model parameters are $\boldsymbol{\theta} = (\mathfrak{S}, [t_0^c, t_1^c], [t_0^t, t_1^t], \tau, \mathbf{m}, \chi, \alpha)$.

The attention matrix $\mathbf{C}(\chi, \mathbf{m})$. A reference $(i, j) \in \mathcal{R}$ transfers one unit of attention from w_i to w_j . On each side, a fraction $(1 - \chi)$ is attributed to the source and χ to the affiliated institutions (weighted by ω_{ik}), so each reference generates four typed edges with weights $(1 - \chi)^2, \chi(1 - \chi), \chi(1 - \chi), \chi^2$ for the SS, SI, IS, II blocks respectively. Summing over $\mathcal{R}$ and applying a binary block-mask $\mathbf{m} = (m_{SS}, m_{SI}, m_{IS}, m_{II}) \in \{0, 1\}^4$, with $m_{SI} = m_{IS}$, gives

$$\mathbf{C}(\chi, \mathbf{m}) = \begin{pmatrix} m_{SS}(1 - \chi)^2 \mathbf{C}_{SS} & m_{SI} \chi(1 - \chi) \mathbf{C}_{SI} \\ m_{IS} \chi(1 - \chi) \mathbf{C}_{IS} & m_{II} \chi^2 \mathbf{C}_{II} \end{pmatrix}. \quad (2)$$

Four masks are bibliometrically meaningful: source-only $(1, 0, 0, 0)$, institution-only $(0, 0, 0, 1)$, bipartite $(0, 1, 1, 0)$, and full-joint $(1, 1, 1, 1)$. The parameter χ controls the source–institution balance in the bipartite and full-joint cases and is inert in the single-mode cases. Block construction details are in the Supplement.

The Galerkin interpretation is $\mathbf{C}(\chi) = \mathbf{B}(\chi)^\top \mathbf{W} \mathbf{B}(\chi)$, a projection of the work-level reference structure onto the unit space, where $\mathbf{B}(\chi)$ is the work-to-unit incidence matrix and χ controls the shape of the basis functions.

Spectral ranking. Let $\mathcal{P} = \mathfrak{S} \cup \mathfrak{I}$ be the *unit* set (?), $N = N_s + N_u$, $r_p = \sum_q C_{pq}$ the outgoing reference flow, $\mathbf{D}_r = \text{diag}(r_p)$, and $\mathbf{A} = \text{diag}(a_p)$ the work-count matrix ($a_p = N_p^s$ for sources, $a_p = M_k$ for institutions). The row-stochastic reference matrix is $\mathbf{H} = \mathbf{D}_r^{-1} \mathbf{C}$. The Katz resolvent (??),

$$\boldsymbol{\pi}^*(\alpha) = (1 - \alpha)(\mathbf{I} - \alpha \mathbf{H}^\top)^{-1} \mathbf{p}, \quad p_p = \frac{a_p}{N_{\text{works}}}, \quad (3)$$

converges for all $\alpha < 1$ and approaches the stationary distribution of the Markov chain with transition matrix $\mathbf{H}$ as $\alpha \rightarrow 1$; ? shows π_q^* equals the long-run expected share of reference flow received by unit q . *Prestige per work* (?) is

$$\mathbf{v}^*(\alpha) = \mathbf{A}^{-1} \boldsymbol{\pi}^*(\alpha), \quad (4)$$

recovered once after convergence of successive substitution on (??). Convergence analysis is in the Supplement. We report $\mathbf{v}^*$ over $\alpha \in \{0.50, 0.75, 0.85, 0.95\}$.

Critical implementation note. Applying $\mathbf{A}^{-1}$ inside the iteration loop iterates on $\mathbf{H}^\top \mathbf{A}$, which is not norm-preserving—this “sums the averages rather than averages the sum” and yields a fixed point with no clean interpretation.

Computation. $\mathbf{C}(\chi, \mathbf{m})$ is sparse ($\text{nnz} \leq 4|\mathcal{R}|$) and is built from a weighted edge list $\mathcal{E}$ accumulated over $\mathcal{R}$, stored in CSR format, and row-normalised to $\mathbf{H}$. The pipeline $\mathcal{R} \rightarrow \mathcal{E} \rightarrow \mathbf{C} \rightarrow \mathbf{H} \rightarrow \boldsymbol{\pi}^* \rightarrow \mathbf{v}^*$ and its parameter entry points are detailed in the Supplement. Implementation uses `scipy.sparse.csr_matrix`.

3 Example: Economics and Commerce

The standing of journals and institutions in economics and commerce (EC, including economics, econometrics, finance, accounting, business, and management) is well reported, offering an opportunity to test dual spectral ranking against established approaches. The broad field also spans relatively unconnected sub-fields (e.g. economics and business) supporting an analysis of the effect of clustering on rankings.

4 Corpus Construction

Corpus construction begins with a discipline-based list of *OpenAlex source entities* (OAS). The list combines entries from three established journal registers: Harzing’s *Journal Quality List* (JQL) ?, the *ERA 2023 Submission Journal List* (SJL) maintained by the Australian Research Council ?, and the *Clarivate Master Journal List* (MJL), drawn from the Web of Science Journal Citation Reports ?. Each entry in these registers includes at least one ISSN identifier together with a text description of its disciplinary coverage.

Because SJL and MJL span all scholarly disciplines, disciplinary filtering is applied before matching. SJL entries are retained where *any* of the up to three assigned *Field of Research* (FoR) codes falls within Division 35 (*Commerce, Management, Tourism and Services*) or Division 38 (*Economics*), encompassing both the two-digit divisional codes and all four-digit group codes within those divisions (3501–3509 and 3801–3803, 3899). Using any FoR column rather than only the primary FoR 1 captures cross-disciplinary journals classified primarily in adjacent fields (e.g., urban planning, engineering, agriculture) that nonetheless carry a secondary or tertiary economics or management classification. MJL entries are retained where the assigned Web of Science category is one of: ECONOMICS, TRANSPORTATION, MANAGEMENT, BUSINESS, or BUSINESS, FINANCE. JQL is not filtered: its scope is already confined to business and economics disciplines. After filtering, the three registers contain 827 (JQL), 2,469 (SJL), and 1,413 (MJL) journals respectively.

4.1 Joining to OpenAlex Sources

Each filtered register is joined independently to the OpenAlex sources snapshot using ISSN as the matching key. For JQL, the single ISSN column is matched against the OpenAlex preferred ISSN (`issn_1`) and the full set of alternative ISSNs stored in the `issn` field. For MJL, both the print ISSN and electronic ISSN fields are tried in sequence. For SJL, all seven ISSN columns (`ISSN 1` through `ISSN 7`) are tried in sequence, stopping at the first hit.

The three matched sets are then unioned to produce the long-list of candidate OAS. A journal appearing in more than one register contributes a single OAS entry, identified by its OpenAlex source identifier. The match statistics for each register are shown in Table ??.

Table 1: Matching of disciplinary journal registers to OpenAlex sources (OAS). Overlap counts are based on shared OpenAlex source identifiers among matched entries.

Register	Journals	Matched to OAS	Match rate	Unmatched
JQL	827	826	99.9%	1
MJL	1,413	1,393	98.6%	20
SJL	2,469	2,193	88.8%	276
<i>Pairwise overlap of matched OAS</i>				
MJL $\cap$ SJL	1,008 sources in common			
MJL $\cap$ JQL	540 sources in common			
SJL $\cap$ JQL	618 sources in common			
MJL $\cap$ SJL $\cap$ JQL	505 sources in all three registers			
Union (long-list)	2,751 unique OAS			

Long-list segment	OAS	Share
MJL only	350	12.7%
SJL only	1,072	39.0%
JQL only	173	6.3%
MJL $\cap$ SJL (not JQL)	503	18.3%
MJL $\cap$ JQL (not SJL)	35	1.3%
SJL $\cap$ JQL (not MJL)	113	4.1%
All three registers	505	18.4%
Total	2,751	100.0%

4.2 Unmatched Journals

The overall unmatched rate is low for JQL (0.1%) and MJL (1.4%) but more substantial for SJL (11.2%). The higher SJL rate reflects the breadth of the ERA register, which deliberately includes lower-profile national and regional journals. Table ?? categorises the 276 unmatched SJL entries.

Table 2: Categorisation of the 276 SJL journals not matched to OpenAlex sources.

Category	Count
Small specialist journal (not indexed in OA)	132
Possible ISSN mismatch (name match found in OA)	47
Regional / national journal	30
Non-English / regional language journal	22
Tax / legal specialist journal	14
Education / pedagogical journal	12
Newsletter / trade publication	9
Working paper / discussion paper series	6
Actuarial / insurance journal	3
Annual edition / book series	1
Total	276

The 47 journals flagged as possible ISSN mismatches were investigated further by fuzzy name-matching against OpenAlex display names. Of these, 42 are high-confidence matches where the ERA-recorded ISSN and the OpenAlex `issn.1` differ (a common occurrence when a register records the print ISSN while OpenAlex uses the electronic ISSN as preferred identifier, or vice versa). A further five cases involve multiple OpenAlex records sharing the same journal name, requiring disambiguation. Three cases are assessed as probable false matches—most notably *Banking and Finance Law Review*, for which the name-similar OpenAlex entry is a distinct American periodical. These 47 candidates are held in a separate inspection table and are not included in the long-list pending editorial review.

The 20 unmatched MJL journals are predominantly regional or non-Anglophone publications (e.g., *Maliye Dergisi*, *Politica Economica*, *Region et Développement*) or have ISSNs not yet registered in the OpenAlex snapshot. The single unmatched JQL entry is the *B.E. Journal of Economic Analysis and Policy*, which is absent from the OpenAlex sources file despite being a peer-reviewed journal.

4.3 FoR Classification Breadth

Within SJL, journals are assigned a primary FoR code at either the two-digit divisional level (35 or 38) or a four-digit group level (e.g., 3501 for *Accounting, Auditing and Accountability*). A two-digit primary code signals that the journal’s scope is too broad to assign to any single group, typically indicating a cross-disciplinary outlet. Of the 2,469 filtered SJL entries, 172 (7.0%) carry a two-digit FoR 1 code. Contrary to the initial hypothesis that breadth might reduce OAS coverage, these cross-disciplinary journals actually exhibit a *lower* unmatched rate (7.0%) than the 4-digit group journals (11.8%). This is consistent with broader-scope journals tending to be larger, more established outlets that OpenAlex is more likely to index. The FoR code granularity is therefore not used as an exclusion criterion, but the two-digit flag is retained as a covariate for downstream scope analysis, since a cross-disciplinary source may yield works spanning multiple research areas beyond the target fields.

4.4 Long-List Composition

The 2,751 OAS in the long-list are distributed across the three registers as shown in Table ?. The SJL-only segment (1,072 sources, 39.0%) is the largest single partition and consists entirely of journals that carry an ERA disciplinary classification but no JCR impact metrics—predominantly emerging, regional, or specialist outlets. The 505 sources appearing in all three registers (18.4%) represent the highest-consensus core: outlets simultaneously recognised by the ERA, classified in JCR, and assessed in Harzing’s quality synthesis. These sources will carry the richest set of quality and bibliometric metadata for subsequent corpus filtering.

4.5 Cross-check against the OpenAlex Discipline-Scoped Journal List

The 2,751 long-list OAS are compiled from disciplinary registers, each of which casts a deliberately broad net: the ERA SJL encompasses any journal with a business or economics secondary classification, the MJL includes transportation and management science outlets, and Harzing’s JQL includes journals that, while valuable references for management researchers, sit primarily in adjacent social-science or natural-science disciplines. To assess whether each long-list entry has substantive disciplinary coverage, the long-list is cross-checked against a second OpenAlex-derived resource—here called the *OpenAlex Journal List* (OJL).

The OJL is constructed from the OpenAlex topics taxonomy. Each source record in the OpenAlex snapshot carries a list of topic assignments derived from the content of its indexed works, with each topic linked to a hierarchical *subfield*, *field*, and *domain*. The OJL retains all OpenAlex sources for which at least one indexed work carries a topic belonging to Field 14 (*Economics, Econometrics and Finance*) or Field 20 (*Business and Management*). This yields 42,756 distinct source entities. For each such source, the OJL records the *in-scope ratio*—the share of the source’s indexed works whose primary topic falls within Fields 14 or 20—as an indicator of disciplinary concentration.

Long-list OAS matched to OJL

Matching is performed by OA source identifier and, where identifiers do not agree between the two snapshots, by shared ISSN (using the OpenAlex preferred ISSN `issn_1` and all alternative ISSNs on both sides). Of the 2,751 long-list OAS, 2,289 (83.2%) are found in the OJL, confirming that a substantial majority of the register-derived list is independently recognised by OpenAlex as having economics or business content. The remaining 462 OAS (16.8%) are absent from the OJL, meaning that OpenAlex has not classified any of their indexed works under Fields 14 or 20.

OJL sources not in the long-list

Conversely, of the 42,756 OJL sources, only 2,289 (5.4%) appear in the disciplinary register long-list. This large residual reflects the heterogeneous composition of OJL: it includes every source for which even a single work is classified in Fields 14 or 20, irrespective of that source’s primary disciplinary focus. Restricting to sources with in-scope ratio above 0.40 yields 9,624 OJL sources absent from the long-list.

These high-ratio OJL sources were examined to understand the gap. The 5,341 OJL sources with in-scope ratio above 0.60 that do not appear in the long-list are categorised in Table ???. The dominant categories are low-output sources (28.6%; fewer than 30 indexed works), journals not represented in any of the three disciplinary registers (27.1%), and non-Anglophone or regional journals unlikely to be captured by the primarily English-language registers (21.8%). A further 12.7% carry no ISSN in OpenAlex and are therefore unmatchable regardless of register coverage. Together these structural reasons account for over 90% of the gap. No further sources are added to the long-list on the basis of the OJL cross-check, as the residual journals are either low in aggregate output, absent from established disciplinary rankings, or lack the identifiers required for reliable provenance tracking.

4.6 Scope Filtering of the Long-list OAS

The 462 long-list OAS absent from the OJL are examined to assess whether their absence reflects a genuine disciplinary mismatch or a structural gap in OpenAlex’s topic coverage. By construction, absence from the OJL implies that OpenAlex has not assigned any work from that source to Fields 14 or 20. This can arise for two substantively different reasons.

First, a journal may have been captured by the disciplinary registers but is primarily housed in an adjacent field—psychology, political science, statistics, or the natural sciences—and appears in the register only because of a secondary or borderline classification. Examples include *Nature Communications*, *American Psychological Review*, and *American Political Science Review*, all of which appear in Harzing’s JQL as reference journals for management researchers but whose indexed content is overwhelmingly outside the target fields.

Second, the OpenAlex topic classifier may not have assigned Fields 14 or 20 to the works of a genuine economics or management journal. This occurs most visibly for the 82 long-list OAS that carry a MJL

Table 3: High in-scope-ratio OJL sources (ratio > 0.60) absent from the register-based long-list, classified by likely reason for absence. In-scope ratio is the fraction of a source’s indexed works whose primary topic falls within Fields 14 or 20 of the OpenAlex taxonomy.

Reason for absence from long-list	Count	Share
Low output (<30 indexed works in OA)	1,528	28.6%
Not in JQL, SJL, or MJL registers	1,447	27.1%
Non-Anglophone / regional journal	1,163	21.8%
No ISSN registered in OpenAlex	676	12.7%
Inactive (last publication before 2010)	239	4.5%
Tax / legal specialist	118	2.2%
Health / medical adjacent	73	1.4%
Conference proceedings	48	0.9%
Working paper series	44	0.8%
Total	5,341	100%

classification in ECONOMICS, TRANSPORTATION, MANAGEMENT, BUSINESS, or BUSINESS, FINANCE: these sources are confirmed in-scope by the Web of Science category system, yet are absent from the OJL because OpenAlex classifies their content primarily under transport science, operations research, or adjacent technology fields rather than Fields 14 or 20. Notable examples include *Transportation Research* parts B, D, and F, *Journal of Transport Geography*, and *Accident Analysis and Prevention*.

Each of the 462 OAS absent from OJL is assigned a scope signal based on register membership and journal name (Table ??).

Table 4: Scope signal assigned to the 462 long-list OAS absent from the OJL, based on register membership and journal name.

Scope signal	Count	Interpretation
WoS-confirmed scope	82	In MJL; absent from OJL due to OA classifier gap, not disciplinary mismatch. Retained.
JQL+SJL borderline	36	In both Harzing and ERA; names ambiguous. Retained pending manual review.
JQL peripheral	79	In JQL only; names suggest adjacent discipline (statistics, IS, sociology). Flagged.
Adjacent field	105	Name clearly outside target fields (natural sciences, medicine, political science, psychology). Flagged.
SJL adjacent	186	In SJL only; captured via a secondary FoR classification in transport, logistics, or engineering. Flagged.
Total	462	

Activity and output criteria

Two further criteria are applied uniformly across all 2,751 long-list OAS, drawing on publication-year data from the OJL for the 2,289 matched sources. A source is flagged *Inactive pre-2000* if its last indexed publication year in OpenAlex is before 2000; such sources cannot contribute works to a modern research corpus. A source is flagged *Very low output* if the ratio of its indexed-work count to the length of its active period (last minus first publication year plus one) falls below 1.0, indicating negligible OpenAlex coverage regardless of whether the journal itself is active. This criterion reflects the practical constraint that work extraction is from OpenAlex: a source with fewer than one indexed work per year will contribute negligibly to the corpus, whether because the journal is genuinely thin or because its content is predominantly behind a paywall and therefore unindexed.

Resulting scope-filtered OAS

Table ?? summarises the `out_of_scope` categorical flag applied to each of the 2,751 long-list OAS. Flag assignment follows a priority order: (i) inactivity before 2000, (ii) very low output rate, (iii) scope signal for OJL-absent sources, and (iv) in-scope otherwise.

Table 5: Distribution of the `out_of_scope` flag across the 2,751 long-list OAS. Sources classified as *In scope* or *WoS-confirmed scope* are retained for work extraction; all others are excluded from corpus assembly.

Flag value	Count	Share	Basis
In scope	2,259	82.1%	OJL-matched; active; adequate OA output
WoS-confirmed scope	82	3.0%	MJL category; absent from OJL (classifier gap)
<i>Excluded:</i>			
SJL adjacent	209	7.6%	SJL-only; transport / logistics / engineering
JQL peripheral	79	2.9%	JQL-only; adjacent social or formal sciences
Adjacent field	59	2.1%	Name-identified non-economics outlet
JQL+SJL borderline	33	1.2%	Borderline; retained for manual review
Very low output	25	0.9%	<1 OA-indexed work per active year
Inactive pre-2000	5	0.2%	No publication after 1999
In scope (net)	2,341	85.1%	
Flagged for exclusion	410	14.9%	
Total long-list	2,751	100%	

The 2,341 in-scope OAS form the *scope-filtered source list* used for work extraction. Of these, 2,259 are confirmed in-scope by the OJL cross-check (in-scope ratio > 0), and 82 are retained on the strength of their Web of Science category despite the absence of a corresponding OJL record. The 33 JQL+SJL borderline sources are included in the count pending manual review; reclassification of any of these as out-of-scope would reduce the list proportionally. The 25 very-low-output sources represent a practical rather than a disciplinary exclusion: their OpenAlex coverage is too sparse to contribute meaningfully to a works-based analysis, but their disciplinary classification is not in question.

Network properties. [Placeholder: reference cross-linking rates, work counts, degree distributions, time-window sensitivity.]

Limitations of the data. [Placeholder: reference cross-link, author disambiguation, institution identity resolution.]

5 Results

Baseline model. [Placeholder.]

Parameter sensitivity. [Placeholder: $\alpha \in \{0.50, 0.75, 0.85, 0.95\}$; $\chi \in \{0.00, 0.25, 0.50, 0.75, 1.00\}$; $\mathbf{m} \in \{(1, 0, 0, 0), (0, 1, 1, 0), (1, 1, 1, 1)\}$; τ sweep.]

Bootstraps. [Placeholder.]

6 Discussion

Face validity. [Placeholder.]

Content validity. [Placeholder.]

Criterion validity — concurrent and predictive. [Placeholder.]

Construct validity. [Placeholder.]

7 Prospects

[Placeholder.]

8 Prospects

[Placeholder.]